

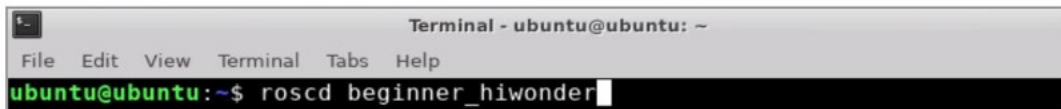
第 10 课 服务数据的自定义与使用

1.自定义服务数据

注意：在自定义服务数据之前，需要先创建工作空间和功能包，具体操作过程可前往目录“第2章 ROS开发入门->ROS基础课程->第3课 创建工作空间与功能包”查看文档。

自定义服务数据的具体操作步骤如下：

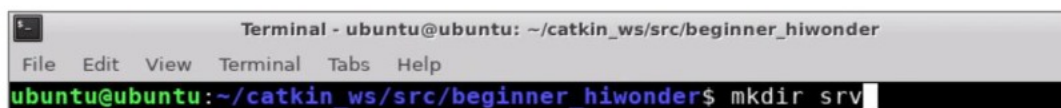
- 1) 打开命令行终端。
- 2) 输入进入功能包所在目录的指令“`roscd beginner_hiwonder`”，并按下回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd beginner_hiwonder
```

注意：若出现提示“`No such package/stack 'beginner_hiwonder'`”，即功能包不存在于环境变量 `ROS_PACKAGE_PATH` 中，具体解决方法可前往目录“第 2 章 ROS 开发入门->ROS 基础课程->第 3 课 创建工作空间与功能包”查看文档，解决此问题后，请重复一次当前步骤的操作。

- 3) 输入指令“`mkdir srv`”，并按下回车，新建用于存放文本文件的文件夹“`srv`”。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir srv
```

- 4) 输入指令“`vi Person.srv`”编辑程序，复制下面程序。如需修改，再按下“`i`”即可修改。修改完成，按下“`Esc`”，输入“`: wq`”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/srv
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/srv$ vi Person.srv
```

```
string name

int8  age

int8  sex
```

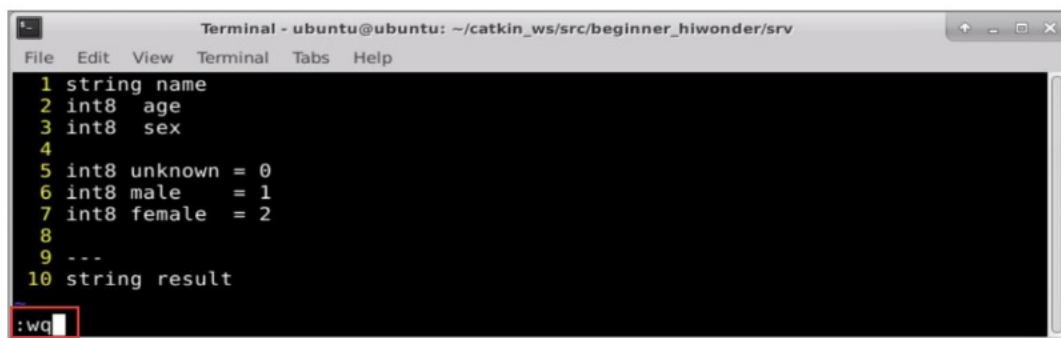
```
int8 unknown = 0
```

```
int8 male    = 1
```

```
int8 female  = 2
```

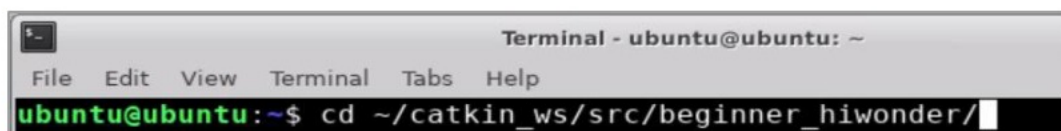
```
---
```

```
string result
```



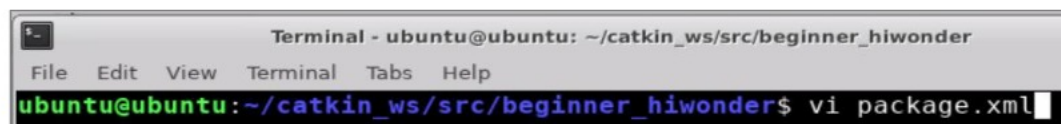
```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/srv
File Edit View Terminal Tabs Help
1 string name
2 int8 age
3 int8 sex
4
5 int8 unknown = 0
6 int8 male    = 1
7 int8 female  = 2
8
9 ---
10 string result
:wq
```

- 5) 输入“`cd ~/catkin_ws/src/beginner_hiwonder/`”，按下回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd ~/catkin_ws/src/beginner_hiwonder/
```

- 6) 输入“`vi package.xml`”，复制下面程序，在下图所示位置添加功能包依赖。如需修改，再按下“`i`”即可修改。修改完成，按下“`Esc`”，输入“`: wq`”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ vi package.xml
```

```
<build_depend>message_generation</build_depend>

<exec_depend>message_runtime</exec_depend>
```



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
49 <!-- Use doc_depend for packages you need only for building documentation:
-->
50 <!-- <doc_depend>doxygen</doc_depend> -->
51 <buildtool_depend>catkin</buildtool_depend>
52 <build_depend>roscpp</build_depend>
53 <build_depend>rospy</build_depend>
54 <build_depend>std_msgs</build_depend>
55 <build_export_depend>roscpp</build_export_depend>
56 <build_export_depend>rospy</build_export_depend>
57 <build_export_depend>std_msgs</build_export_depend>
58 <exec_depend>roscpp</exec_depend>
59 <exec_depend>rospy</exec_depend>
60 <exec_depend>std_msgs</exec_depend>
61 <build_depend>message_generation</build_depend>
62 <exec_depend>message_runtime</exec_depend>
63
64
65 <!-- The export tag contains other, unspecified, tags -->
66 <export>
67 <!-- Other tools can request additional information be placed here -->
68
69 </export>
70 </package>
:wq
```

- 7) 输入“vi CMakeLists.txt”，再按下“i”，修改“CMakeLists.txt”文件。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ vi CMakeLists.txt
```

- 8) 在下图所示位置添加所需的编译选项“message_generation”。

```
0
7 ## Find catkin macros and libraries
8 ## if COMPONENTS list like find_package(catkin REQUIRED COMPONENTS xyz)
9 ## is used, also find other catkin packages
10 find_package(catkin REQUIRED COMPONENTS
11   roscpp
12   rospy
13   std_msgs
14   message_generation
15 )
```

- 9) 找到下图所示代码，将红框部分进行反注释，并添加所需的编译选项“Person.srv”。

```
53 #   Message1.msg
54 #   Message2.msg
55 # )
56
57 ## Generate services in the 'srv' folder
58 add_service_files(
59   FILES
60   Person.srv
61 )
62
63 ## Generate actions in the 'action' folder
64 # add_action_files(
```

10) 找到下图所示代码，将红框部分进行反注释，确保所需的编译选项生效。

```
67 #   Action2.action
68 # )
69
70 ## Generate added messages and services with any dependencies listed here
71 generate_messages(
72     DEPENDENCIES
73     std_msgs
74 )
75
76 #####
77 ## Declare ROS dynamic reconfigure parameters ##
78 #####
```

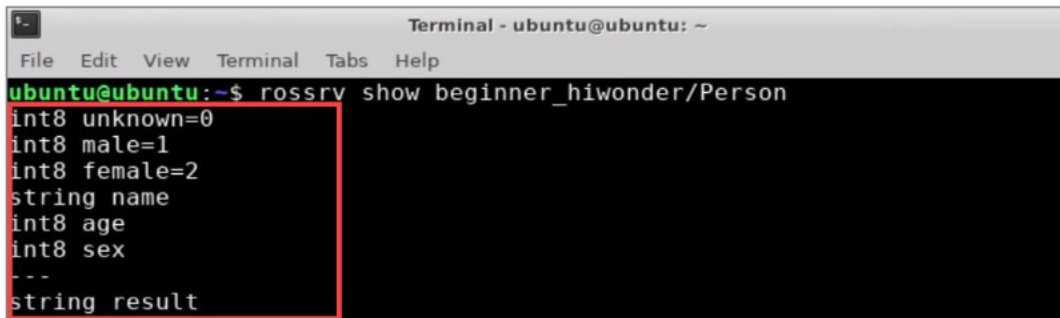
11) 找到下图所示代码，将红框部分进行反注释，并添加所需的编译选项“message_runtime”。

```
lso need
103 ## CATKIN_DEPENDS: catkin_packages dependent projects also need
104 ## DEPENDS: system dependencies of this project that dependent projects also
    need
105 catkin_package(
106 #   INCLUDE_DIRS include
107 #   LIBRARIES beginner_hiwonder
108 CATKIN_DEPENDS roscpp rospy std_msgs message_runtime
109 #   DEPENDS system_lib
110 )
```

12) 修改完成，按下“Esc”，输入“: wq”保存并退出。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
184 #   FILES_MATCHING PATTERN "*.h"
185 #   PATTERN ".svn" EXCLUDE
186 # )
187
188 ## Mark other files for installation (e.g. launch and bag files, etc.)
189 # install(FILES
190 #   # myfile1
191 #   # myfile2
192 #   DESTINATION ${CATKIN_PACKAGE_SHARE_DESTINATION}
193 # )
194
195 #####
196 ## Testing ##
197 #####
198
199 ## Add gtest based cpp test target and link libraries
200 # catkin_add_gtest(${PROJECT_NAME}-test test/test_beginner_hiwonder.cpp)
201 # if(TARGET ${PROJECT_NAME}-test)
202 #   target_link_libraries(${PROJECT_NAME}-test ${PROJECT_NAME})
203 # endif()
204
205 ## Add folders to be run by python nosetests
206 # catkin_add_nosetests(test)
:wq
```

13) 输入指令“**rossrv show beginner_hiwonder/Person**”，并按下回车，查看写入的消息字段能否被系统识别。当出现下图红框所示字样，即代表识别成功。

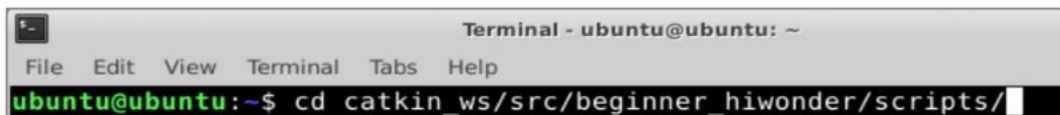


```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rossrv show beginner_hiwonder/Person
int8 unknown=0
int8 male=1
int8 female=2
string name
int8 age
int8 sex
---
string result
```

2.服务数据的使用

2.1 创建服务器和客户端代码

- 1) 打开Linux命令行终端。
- 2) 输入指令“**cd catkin_ws/src/beginner_hiwonder/scripts/**”，并按下回车，进入用于存放Python脚本的文件夹“**scripts**”。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/beginner_hiwonder/scripts/
```

- 3) 输入指令“**vi person_server.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi person_server.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将执行/show_person 服务，服务数据类型 beginner_hiwonder::Person
```



```
import rospy

from beginner_hiwonder.srv import Person, PersonResponse

def personCallback(req):

    # 显示请求数据

    rospy.loginfo("Person: name:%s age:%d sex:%d", req.name, req.age, req.sex)

    # 反馈数据

    return PersonResponse("OK")

def person_server():

    # ROS 节点初始化

    rospy.init_node('person_server')

    # 创建一个名为/show_person 的 server，注册回调函数 personCallback

    s = rospy.Service('/show_person', Person, personCallback)

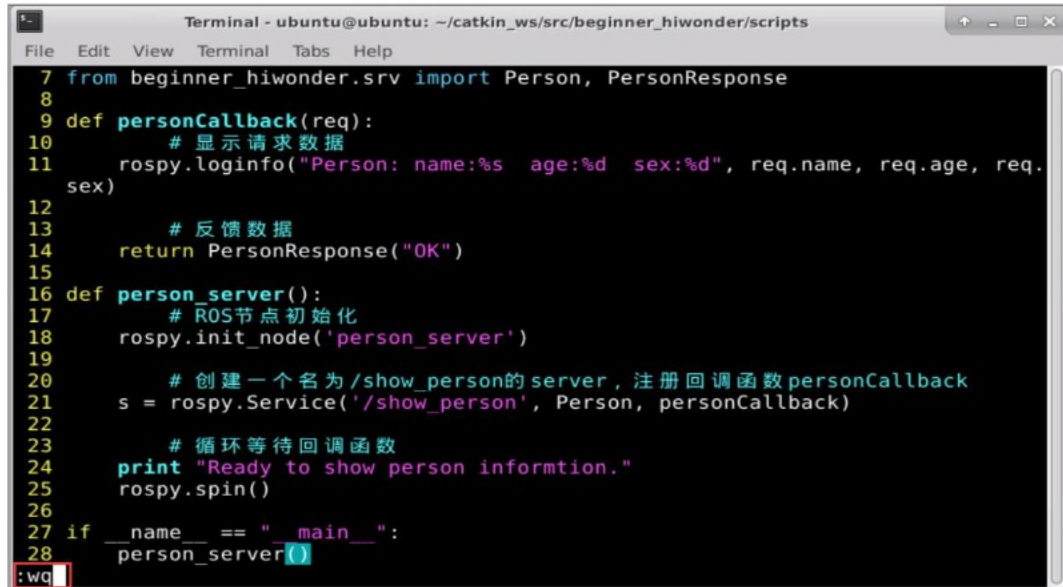
    # 循环等待回调函数

    print "Ready to show person informtion."

    rospy.spin()
```

```
if __name__ == "__main__":
```

```
    person_server()
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
7 from beginner_hiwonder.srv import Person, PersonResponse
8
9 def personCallback(req):
10     # 显示请求数据
11     rospy.loginfo("Person: name:%s age:%d sex:%d", req.name, req.age, req.
sex)
12
13     # 反馈数据
14     return PersonResponse("OK")
15
16 def person_server():
17     # ROS节点初始化
18     rospy.init_node('person_server')
19
20     # 创建一个名为/show_person的server, 注册回调函数 personCallback
21     s = rospy.Service('/show_person', Person, personCallback)
22
23     # 循环等待回调函数
24     print "Ready to show person informtion."
25     rospy.spin()
26
27 if __name__ == "__main__":
28     person_server()
:wq
```

4) 输入指令“**vi person_client.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi person_client.py
```

```
#!/usr/bin/env python
```

```
# -*- coding: utf-8 -*-
```

```
# 该例程将请求/show_person 服务，服务数据类型 beginner_hiwonder::Person
```

```
import sys
```

```
import rospy
```

```
from beginner_hiwonder.srv import Person, PersonRequest
```



```
def person_client():

    # ROS 节点初始化

    rospy.init_node('person_client')

    # 发现/spawn 服务后，创建一个服务客户端，连接名为/spawn 的 service

    rospy.wait_for_service('/show_person')

    try:

        person_client = rospy.ServiceProxy('/show_person', Person)

        # 请求服务调用，输入请求数据

        response = person_client("Tom", 20, PersonRequest.male)

        return response.result

    except rospy.ServiceException, e:

        print "Service call failed: %s"%e

if __name__ == "__main__":

    #服务调用并显示调用结果

    print "Show person result : %s" %(person_client())
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
5
6 import sys
7 import rospy
8 from beginner_hiwonder.srv import Person, PersonRequest
9
10 def person_client():
11     # ROS节点初始化
12     rospy.init_node('person_client')
13
14     # 发现 /spawn服务后，创建一个服务客户端，连接名为 /spawn的 service
15     rospy.wait_for_service('/show_person')
16     try:
17         person_client = rospy.ServiceProxy('/show_person', Person)
18
19         # 请求服务调用，输入请求数据
20         response = person_client("Tom", 20, PersonRequest.male)
21         return response.result
22     except rospy.ServiceException, e:
23         print "Service call failed: %s"%e
24
25 if __name__ == "__main__":
26     #服务调用并显示调用结果
27     print "Show person result : %s" %(person_client())
:wq
```

5) 输入指令“**chmod +x person_server.py**”和“**chmod +x person_client.py**”，并按下回车，赋予文件可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x person_server.py

Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x person_client.py
```

2.2 运行服务器和客户端节点

1) 输入指令“**cd ~/catkin_ws**”，并按下回车，进入catkin工作空间。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ cd ~/catkin_ws
```

2) 输入指令“**catkin_make**”，并按下回车，构建目录中所有的功能包。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ catkin_make
```

- 3) 输入指令“**source ./devel/setup.bash**”，并按下回车，刷新工作空间的环境。



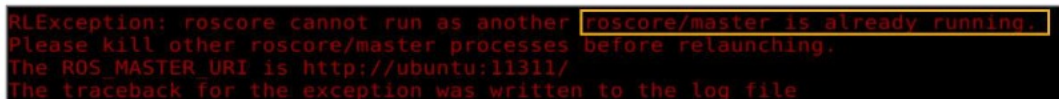
```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ source ./devel/setup.bash
```

- 4) 输入指令“**roscore**”，并按下回车，启动节点管理器。



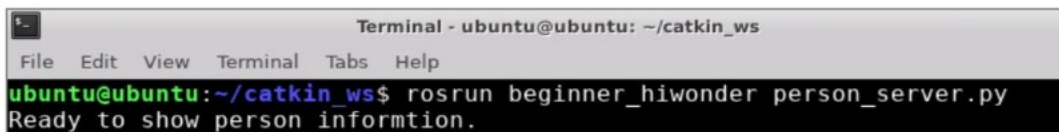
```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ roscore
```

若已开启，则会出现以下提示：



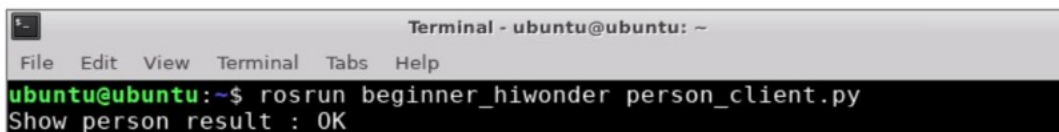
```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

- 5) 输入指令“**roslaunch beginner_hiwonder person_server.py**”，并按下回车，运行服务器节点。如需停止节点的运行，可按下快捷键“**Ctrl+C**”。



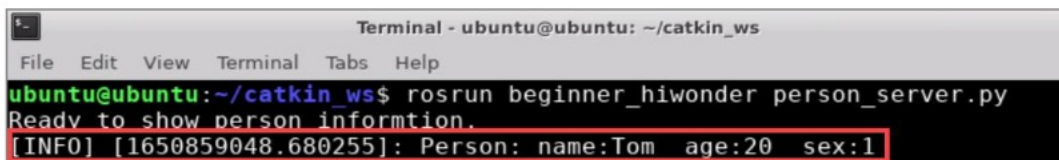
```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ roslaunch beginner_hiwonder person_server.py
Ready to show person informtion.
```

- 6) 打开一个新的命令行终端，输入指令“**roslaunch beginner_hiwonder person_client.py**”，并按下回车，运行客户端节点。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch beginner_hiwonder person_client.py
Show person result : OK
```

- 7) 运行客户端节点后，启动服务器节点的终端窗口会打印下图红框所示内容。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ roslaunch beginner_hiwonder person_server.py
Ready to show person informtion.
[INFO] [1650859048.680255]: Person: name:Tom age:20 sex:1
```